

Binary Stroke Prediction

Stage 1: CRISP-DM Methodology Implementation

 Team: P3 - T11

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 Colab Notebook: [group11_lab3_stage1.ipynb](#)

Business Understanding

🎯 Problem Objective

We aim to develop a binary classification model to predict whether an individual is at risk of experiencing a stroke based on their demographic, lifestyle, and clinical characteristics.

Target Variable : Stroke occurrence (1 = stroke occurred, 0 = no stroke)

Task Type : Binary Classification (Supervised Learning)

💡 Decision Context

Healthcare Impact:

- Strokes are the second leading cause of death globally, accounting for approximately 11% of total deaths (WHO, 2022)

Preventive Care Value:

Proactive identification allows for targeted screening programs and patient education

- Cost-effective prevention is preferable to expensive emergency treatment and long-term rehabilitation

Decision Context & Use Cases:

Context : Routine health assessments at primary care facilities, annual check-ups, or health screening programs

Decision Point : Should this patient be:

1. Referred for comprehensive cardiovascular assessment?
2. Enrolled in a stroke prevention program?
3. Scheduled for more frequent monitoring?
4. Provided with targeted lifestyle intervention recommendations?

⚠️ Assumptions & Risks

Data & Modelling Assumptions:

- **Binary Classification Sufficiency** : Risk can be meaningfully categorized as high/low (rather than requiring multiple risk levels)
- **Representativeness** : Dataset represents the target population for screening (though we acknowledge potential sampling bias)
- **Generalizability** : Patterns learned from the dataset will generalize to similar populations

Class Imbalance Risk (High Impact)

- **Issue** : Stroke cases are significantly rarer than non-stroke cases in the population
- **Consequence** : Models may achieve high accuracy by simply predicting "no stroke" for everyone.
- Consider stratified sampling in train/test split

False Negative Risk (High Impact)

- **Issue** : Failing to identify someone who will have a stroke (missed detection)
- **Consequence** : Delayed intervention, potential adverse outcomes for patients

False Positive Risk (Medium Impact)

- **Issue** : Incorrectly flagging someone as high-risk
- **Consequence** : Unnecessary anxiety, healthcare resource consumption, potential overtreatment

Decision Process: Patient Assessment → Risk Prediction → Risk Stratification → Intervention Decision

Data Understanding: Dataset Overview

Data Source

Source: [Stroke Prediction Dataset](#)

Platform: Kaggle

Accessibility: Public

Dataset Dimensions

5,110

Records

12

Features

Target Definition

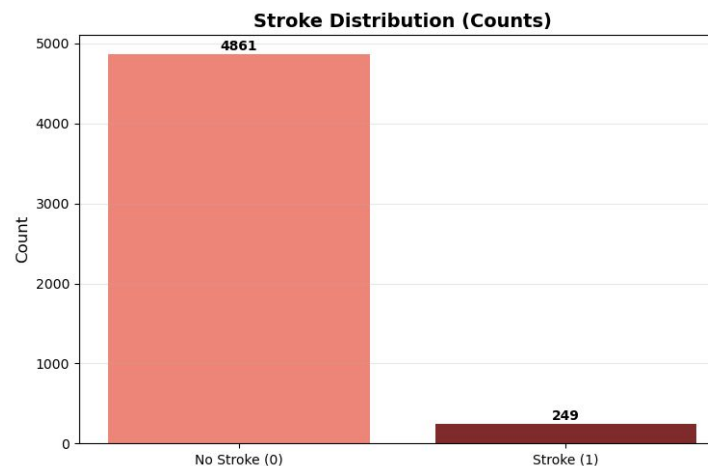
Target Variable: stroke

Task Type: **Classification**

Numerical Features

- age
- bmi
- avg_glucose_level

Data Distribution Overview



Stroke Distribution (Percentage)

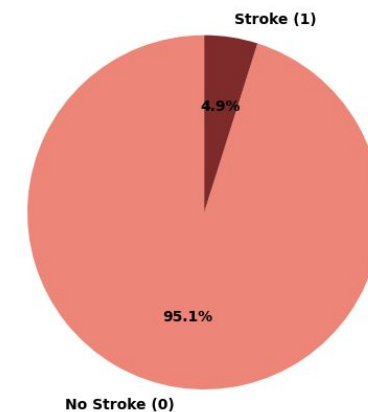


Figure 1: Distribution of stroke outcomes in the dataset, highlighting severe class imbalance (95.1% no stroke vs. 4.9% stroke).

Categorical Features

Nominal: gender, hypertension, heart_disease, ever_married, work_type, residence_type

Ordinal: smoking_status

Note: id is considered an identifier and has been excluded in this overview.

Data Understanding: Quality Assessment

🔍 Data Quality Issues

● Missing Values:

3.93% of records have missing values in `bmi`

● Outliers

43 extreme values (`age < 1`) detected in `age`

110 extreme values (`bmi > 60`) detected in `bmi`

627 extreme values (right skew) detected in `avg_glucose_level`

● Inconsistencies:

- 'Unknown' in `smoking_status` indicates incomplete data collection.
- Imbalanced categories may affect model training.
- Higher stroke rates present in missing `bmi` group.

⚠️ Data Leakage Risks

Potential Leakage Sources:

- Transformations performed before train-test split, causing test data statistics to influence training.
- Oversampling applied before splitting, leading to synthetic samples derived from test data.

Mitigation: All preprocessing will be performed after train-test splitting and fitted only on the training data.

✅ Assumptions

Data Collection Assumptions:

- Patient records are accurately documented and reflect true health status
- Available features capture major known stroke risk factors
- No systematic measurement or recording bias significantly distorts variables

Temporal Assumptions:

- Feature to stroke risk relationships remain relatively stable over time
- Current features adequately represent current risk only
- No major shifts in population health trends/clinical practices

🚫 Data Limitations

What the data does NOT represent:

- Missing clinical factors like genetics, medications, detailed lifestyle, family history
- Dataset represents a static snapshot, not health trajectory

⚖️ Representativeness

Dataset may not generalise beyond source population due to geographic and cultural specificity. Missing values in BMI (~3.9%) and `smoking_status` (~30% "Unknown") suggest possible selection bias in data collection. No systematic measurement or recording bias significantly distorts variables.

Data Preparation

Dataset cleaning and feature engineering steps

Data Cleaning

Missing Values Handling:

Strategy: Median imputation + missing flag
Feature(s) affected: bmi

Outlier Treatment:

- **age:** Remove records with age < 1 year (likely data entry error)
- **gender:** Remove 'Other' gender it is extremely rare within dataset (only 1 record).
- **bmi:** Retained as extreme values are medically plausible (e.g., severe obesity)
- **avg_glucose_level:** Retained as high values valid for diabetic patients (up to ~270 mg/dL)

Categorical Encoding

Encoding Method:

One-hot encoding (nominal features, drop='if_binary')

Features Encoded:

- **gender:** 1 column
- **residence_type:** 1 column
- **ever_married:** 1 column
- **work_type:** 5 columns

Encoding Method:

Ordinal encoding (ordinal categorical)

Features Encoded:

- **smoking_status:**
 - -1: Unknown
 - 0: Never smoked
 - 1: Formerly smoked
 - 2: Smokes
- 'Unknown' values flagged separately via **smoking_unknown_flag**

Train/Test Split

Split Configuration:

[80%]

Training

[20%]

Testing

Split Parameters:

- Random state: 42
- Stratified: Yes

Training set: 4,052 records | **Test set:** 1,014 records

Resampling

Method: SMOTE-NC

Rationale: Addresses severe class imbalance (~5% stroke) by generating synthetic minority samples whilst appropriately handling categorical features.

After resampling

Training set : 5,779 records | **Test set :** (unchanged)

Baseline Modelling

Dummy baseline + Classical models with default settings only

Dummy Classifier

Strategy: Stratified (generates random predictions by respecting the training set's class distribution)

Purpose: Establishes the minimum baseline performance that any valid model must outperform.

Rationale: Because the dataset is severely imbalanced (95% negative), a stratified dummy prevents the baseline from artificially scoring on metrics like Recall and F1-Score. It acts as a strict sanity check to prove our real models are genuinely learning clinical patterns, not just guessing based on probability.

Model 1: Logistic Regression

Algorithm: Logistic Regression

Configuration: Default parameters (no tuning)

Suitability: Interpretable linear model well-suited for binary classification; coefficients indicate feature importance; handles scaled features effectively; robust baseline for medical risk prediction.

Parameters:

- Penalty = 'l2'
- c = 1.0
- l1_ratio = 0.0
- class_weight = None

Model 2: Decision Tree

Algorithm: Decision Tree Classifier

Configuration: Default parameters (no tuning)

Suitability: Captures non-linear relationships and feature interactions; does not require feature scaling; provides interpretable decision rules for clinical use; complements linear assumptions of Logistic Regression.

Stage 1 Constraint: No hyperparameter tuning, dimensionality reduction, or ensemble methods.

Results: Baseline Comparison

Performance Metrics Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Dummy (stratified)	0.66	0.04	0.24	0.06	0.50
Logistic Regression★	0.84	0.17	0.56	0.26	0.81
Decision Tree	0.87	0.11	0.24	0.15	0.57

Table 1: Test set performance metrics for all initial models. ★ indicates best performer.

Model	Train Recall	Test Recall	Gap
Logistic Regression	0.70	0.56	~0.14
Decision Tree	1.00	0.24	~0.76

Table 2: Train - test recall comparison and overfitting assessment for initial models.

Key Observations

- Logistic Regression is the best performer (ROC-AUC: 0.81, Recall: 0.56), outperforming the dummy baseline by 31% and 18% respectively.
- Decision Tree shows severe overfitting (train recall 1.00 vs. test recall 0.24), whilst Logistic Regression generalises well (train-test gap ~0.14).
- Test recall of 0.56 means 44% of at-risk patients would be missed, requiring further refinement for clinical deployment.

Performance Visualisation

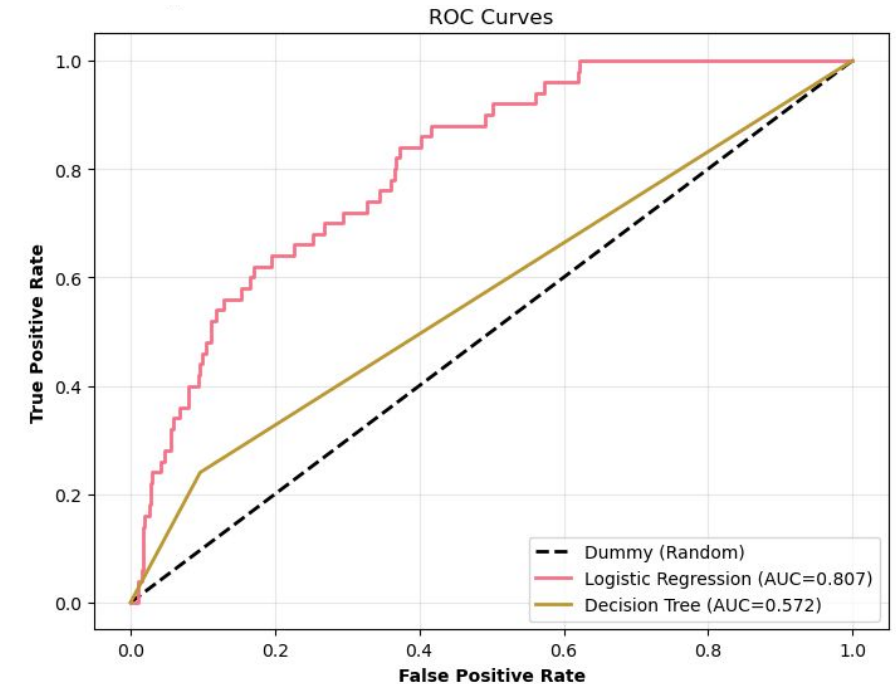


Figure 2: ROC curves comparing model performance on the test set

The ROC curve demonstrates that Logistic Regression substantially outperforms both the Dummy Classifier and Decision Tree. Logistic Regression's curve occupies the upper-left region of the plot, indicating superior true positive rate at acceptable false positive rates, whilst the Decision Tree's performance only marginally exceeds random classification.

Initial Evaluation

🔗 Sanity Checks

✓ Model Behavior Validation:

- All three models converge without errors
- Predictions are binary and within valid probability range (0 – 1)
- No NaN or infinite values detected

✓ Performance Consistency:

- Logistic Regression test metrics are plausible given SMOTE resampled data
- Decision Tree shows train-test gap, indicating overfitting
- Results reproducible with fixed `random_state = 42`

✓ Baseline Comparison:

- Both models beat Dummy on F1 and ROC-AUC
- Performance ordering is logical: Logistic Regression > Decision Tree > Dummy on key metrics (recall, AUC)

📌 Feasibility Assessment

Against Success Criteria:

- Recall >60%: Logistic Regression achieves 56% (close but below target)
- ROC-AUC >0.70: Logistic Regression achieves 0.81 (exceeds target)
- Precision >40%: Best is Logistic Regression with 17% (well below target)

Feasibility Verdict:

- Yes, ROC-AUC substantially exceeds target, demonstrating strong discriminative ability. Recall is within reach and addressable through threshold tuning and hyperparameter optimisation in Stage 2.

⚠️ Concerning Patterns

Issues Identified:

- Low precision across all models (~17% best), meaning ~5 false alarms per true positive
- Decision Tree recall (24%) matches Dummy, meaning no effective minority-class learning

Unexpected Behaviors:

- Decision Tree overfits severely (depth 31, 592 leaves) yet underperforms Logistic Regression on every metric except accuracy
- `bmi_missing` flag and `ever_married` rank among top Logistic Regression coefficients, possibly acting as proxies for age or healthcare engagement rather than direct causal factors

🧠 Learning Assessment

Are models learning meaningful patterns?

Logistic Regression shows genuine learning: AUC improves from 0.50 to 0.81 (+31%) and recall from 24% to 56% (+32%), with top predictors (age, glucose, cardiovascular conditions) aligning with known clinical risk factors. Decision Tree shows limited generalisation: recall matches Dummy at 24% and AUC is only 0.57, suggesting severe overfitting (depth 31) rather than meaningful pattern learning.

Conclusion: Logistic Regression demonstrates meaningful learning with +31% AUC and +32% recall improvement over random baseline. Decision Tree does not generalise.

Next Steps: Stage 2 Plan

🔧 Feature Engineering

Planned Techniques:

Age-based risk categories

- Create age groups: <40, 40-60, 60-75, 75+
- Medical literature shows non-linear stroke risk progression with age

Interaction features

- **hypertension × age**: Combined effect of hypertension in older patients
- **glucose × BMI**: Relationship between metabolic factors

⚙️ Hyperparameter Tuning

Approach:

- **Stratified K-Fold (K=5 or 10)** - Balanced stroke/non-stroke distribution.
- **Grid Search** - Systematic hyperparameter tuning.
- **Random Search** - Efficient exploration of parameter space.
- **Model complexity control (Decision Tree)** - Limit depth and splits to prevent overfitting.

Parameters to Tune:

Logistic Regression

- **C**: Regularization strength → [0.001, 0.01, 0.1, 1, 10, 100]
- **penalty**: ['l1', 'l2']
- **solver**: ['liblinear', 'saga']
- **class_weight**: [None, 'balanced']

Decision Tree

- **max_depth**
- **min_samples_split**
- **min_samples_leaf**
- **criterion**: ['gini', 'entropy']

⊕ Model Interpretability

Global Interpretability:

- **Feature importance rankings**: Identify most influential predictors across all patients
- **Partial dependence plots**: Visualise individual feature effects on stroke probability

Local Interpretability:

- **Explain individual patient predictions**
- **Show how features contribute to specific high-risk cases**

References and Tools

References

Dataset Source:

<https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset/data>

Supplementary Materials

Method/ Concept	Purpose	References
Q - Q Plot	Assesses whether data follows a theoretical distribution (e.g., normal distribution).	https://www.datacamp.com/tutorial/q-q-plot/
Chi-Square Test / Cramér's V	Tests association between categorical variables; Cramér's V measures strength of association.	https://www.scribbr.com/statistics/chi-square-tests/ https://www.ibm.com/docs/en/cognos-analytics/12.0.x?topic=terms-cramers-v
Student's t-test	Compares the means of two groups to determine whether they are statistically different.	https://en.wikipedia.org/wiki/Student's_t-test
Missingness Analysis	Examines patterns and mechanisms of missing data (MCAR, MAR, MNAR).	https://www.sciencedirect.com/topics/mathematics/missingness
Pearson Correlation	Measures the strength and direction of linear relationship between continuous variables.	https://www.sciencedirect.com/topics/computer-science/pearson-correlation
SMOTENC	Oversampling method for handling imbalanced datasets with categorical features.	https://imbalanced-learn.org/stable/references/generated/imblearn.over-sampling.SMOTENC.html

Tools and Libraries

Programming Language: Python

Core Libraries:

Library	Version	Purpose
pandas	[3.0.0]	Data manipulation
numpy	[2.4.2]	Numerical computing
scikit-learn	[1.8.0]	ML algorithms
matplotlib	[3.10.8]	Visualization
seaborn	[0.13.2]	Statistical plots

Development Environment:

Jupyter Notebook

Note: All code and experiments are reproducible with the specified versions and random seeds.

Colab Notebook:

<https://colab.research.google.com/drive/1AlCvbf8TXUzEd4PsgAd4AERQz1Haaqym?usp=sharing>

GitHub Repository:

<https://github.com/timothyckl/INF2008-P3-II-StrokePrediction>



TEAM CONTRIBUTIONS

Individual Contributions

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Individual Contributions:

- Led business understanding research, data understanding documentation.
- Worked on slides.

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